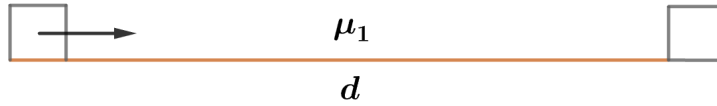


2024 F=ma Exam: Problem 9

Kevin S. Huang



The energy dissipated by friction on the first track is

$$E_d = f_1 d = \mu_1 m g d$$

The energy dissipated by friction on the second track is

$$E_d = f_1(d/2) + f_2 x = \frac{\mu_1 m g d}{2} + \mu_2 m g x$$

Since these are the same,

$$\mu_1 m g d = \frac{\mu_1 m g d}{2} + \mu_2 m g x$$

$$\frac{\mu_1 d}{2} = \mu_2 x$$

$$x = \frac{\mu_1 d}{2\mu_2}$$

The total distance on the second track is

$$D = \frac{d}{2} + x = \frac{d}{2} \left(1 + \frac{\mu_1}{\mu_2} \right) = \frac{100 \text{ m}}{2} \left(1 + \frac{0.8}{0.2} \right) = 250 \text{ m}$$

so the answer is C.